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Capabilities of rule representations for automated compliance checking in healthcare buildings ¹,Zijing Zhang; Nicholas Nisbet; Ling Ma; Tim Broyd ¹, 2023 ¹,10.1016/j.autcon.2022.104688 ¹,"A suitable rule representation is essential to enable automated compliance checking of building design... This paper aims to identify capabilities needed for rule representation by using healthcare design regulations as an example... The results showed 18 capabilities for healthcare rule representation, where 16 are required, and two are desirable ².",Automation in Construction ¹

Automated rule-based safety inspection and compliance checking of temporary guardrail systems in construction ³,K. W. Johansen; J. Teizer; C. Schultz ³,2024 ³,10.1016/j.autcon.2024.105849 ³,This paper proposes an automated rule-based inspection and compliance-checking system that can assist in detecting faulty guardrails in live work environments. The classification approach utilizes safety design and mimics the steps of human guardrailing compliance assessment... This first-of-a-kind classification approach is reliable and scalable and successfully classifies various scenarios of guardrail systems for fall protection ⁴.,Automation in Construction ³

Rule capture of automated compliance checking of building requirements: a review ⁵,Zijing Zhang; Ling Ma; Tim Broyd ⁵,2023 ⁵,10.1680/jsmic.23.00005 ⁵,"Automated compliance checking (ACC) was proposed as a solution to the error-prone and time-consuming manual compliance-checking process. Rule capture is a crucial bottleneck of ACC. This paper conducts a systematic literature review and identifies challenges in rule capture for ACC... The findings reveal that existing representations cannot handle 'unknowns' and 'side effects', lack ability to deal with ambiguous rules, and are often restricted by the rule engine or data model ⁶.",Proc. ICE - Smart Infrastructure and Construction ⁵

CODE-ACCORD: A Corpus of building regulatory data for rule generation towards automatic compliance checking ⁷,Hansi Hettiarachchi; Amna Dridi; Mohamed M. Gaber; Pouyan Parsafard; Nicoleta Bocaneala; Katja Breitenfelder; Gonal Costa; Maria M. Hedblom; Mihaela Juganaru-Mathieu; Thamer Mecharnia; Sumee Park; He Tan; Abdel-Rahman H. Tawil; Edlira Vakaj ⁸,2025 ⁹,10.1038/s41597-024-04320-x ¹⁰,"Automatic Compliance Checking (ACC) necessitates automating the interpretation of building regulations. We introduce CODE-ACCORD, a dataset of 862 sentences from building regulations of England and Finland. Each sentence was manually annotated with entities and relations to facilitate machine-readable rule generation. The final dataset comprises thousands of entities and relations, serving as a robust ground truth for applying ML and NLP techniques to ACC ¹¹ ¹².",Scientific Data ⁹

Moving automated compliance checking to the operational phase of the building life-cycle: analysis and feasibility study in the UK ¹³,Thomas Beach; Jonathan Yeung; Ali Ghorghi; Yacine Rezgui ¹⁴,2024 ¹³, 10.1080/15623599.2024.2366727 ¹⁵,"Construction regulations ensure baseline safety for built assets. Formal compliance checks occur at defined life-cycle points and often lack automation. This paper extends compliance checking into the operational phase of buildings using automated/semi-automated data

capture and analysis. Findings show that whole-life compliance checking is vital but impractical through purely manual inspections, and identify viable use cases and technologies for an automated approach ¹⁶ ¹⁷ .", International Journal of Construction Management ¹⁵ ¹⁸

A SHACL-Based Approach for Enhancing Automated Compliance Checking with RDF Data ¹⁹ , Joseph Anim; Livio Robaldo; Adam Z. Wyner ²⁰ , 2024 ²¹ , 10.3390/info15120759 ²¹ , "Automated Compliance Checking (ACC) is increasingly important for enforcing legal regulations. This paper leverages Semantic Web technologies, focusing on executing SHACL-SPARQL rules on RDF data. The approach addresses limitations of traditional ACC by handling nuanced temporal and aggregate requirements in legal reasoning. A case study on regulations with aggregate and temporal facets demonstrates the effectiveness of the proposed solution in improving ACC via standardized, interoperable Semantic Web methods ²² ²³ .", Information (MDPI) ²¹

"Object-centred automated compliance checking: a novel, bottom-up approach ²⁴ ", Omar Doukari; David Greenwood; Kay Rogage; Mohamad Kassem ²⁵ , 2022 ²⁶ , 10.36680/j.itcon.2022.017 ²⁷ , "Building Information Modelling (BIM) represents built assets as objects with identity, attributes, and relations, enabling new compliance approaches. This paper proposes a bottom-up object-centred method for automated model checking, using a simulated annealing-inspired classification. A prototype demonstrates four key processes (content validation, model completeness, smart object, and design option checking) and satisfies desired compliance checking characteristics. Case studies show the approach saves around 125 minutes per object compared to manual methods ²⁸ ²⁹ .", Journal of IT in Construction (ITcon) ³⁰

Automated code compliance checking research based on BIM and knowledge graph ³¹ , Junlong Peng; Xiangjun Liu ³² , 2023 ³² , 10.1038/s41598-023-34342-1 ³³ , "Traditional drawing reviews rely on experts, but this study achieves automatic code compliance checking by combining BIM and knowledge graphs. Regulatory text is converted into a computer-readable structured language using NLP to form a human-machine-readable knowledge graph. BIM model information is extracted and mapped to compliance rules. An example validates that this method effectively addresses the "manual dependency" and "inefficiency" issues in the review process ³⁴ ³⁵ .", Scientific Reports ³¹

Knowledge-Informed Semantic Alignment and Rule Interpretation for Automated Compliance Checking ³⁶ , Z. Zheng; Y. C. Zhou; X. Z. Lu; J. R. Lin ³⁶ , 2022 ³⁶ , 10.1016/j.autcon.2022.104524 ³⁷ , "Automated rule checking (ARC) in construction requires intelligent rule interpretation from regulatory text and alignment of concepts across models and regulations. This paper presents a knowledge-informed framework using NLP to bridge semantic gaps between design BIM models and textual regulations. An ontology captures domain knowledge (concepts, synonyms, constraints). Semantic alignment and conflict resolution techniques, aided by domain knowledge and unsupervised learning, enhance rule interpretation. The method generates SPARQL queries for model checking, enabling interpretation of complex rules with implicit data. Experiments show ~90% accuracy, significantly outperforming keyword matching, and the automated rule interpretation is ~5× faster than manual expert interpretation ³⁸ ³⁹ .", Automation in Construction ³⁶

Automated Compliance Checking System for Structural Design Codes in a BIM Environment ⁴⁰ , Wonhui Goh; Jaeguk Jang; Sang I. Park; Bong-Hyuck Choi; Heuiseok Lim; Goangseup Zi ⁴¹ , 2024 ⁴⁰ , 10.1007/s12205-024-1121-5 ⁴² , "This

study proposes a framework to develop an ACC system for verifying structural calculations in openBIM format. The framework includes: (1) classifying design codes by structural component and limit state, (2) preparing a structural model in IFC, (3) interpreting design code conditions into a machine-readable rule language, and (4) designing an ACC system focusing on rule flow in structural codes. A detailed example uses ifcOWL, SWRL, and a Drools engine to check a prestressed concrete bridge design. The ACC system successfully verified the design example. The framework can be adapted for various codes and user requirements in structural engineering ⁴³ ⁴⁴.", KSCE Journal of Civil Engineering ⁴⁵ ⁴⁰

NLP-based Automated Compliance Checking of Data Processing Agreements against GDPR ⁴⁶, Orlando Amaral Cejas; Muhammad I. Azeem; Sallam Abualhaija; Lionel Briand ⁴⁷ ⁴⁸, 2023 ⁴⁹, 10.1109/TSE.2023.3288901 ⁵⁰,"When a data processor differs from the data controller, GDPR mandates a Data Processing Agreement (DPA). Ensuring a DPA complies with GDPR is challenging due to the effort in mapping GDPR requirements to the DPA. This paper presents an automated solution using NLP to check a given DPA against GDPR "shall" requirements. It extracts requirements from GDPR and creates a glossary of legal concepts, then automatically generates representations of the DPA text and compares them to the GDPR requirements. The approach detects non-compliance and suggests missing information in the DPA. In experiments on 30 real DPAs, it achieved ~89% precision, ~82% recall, and ~84% accuracy, improving accuracy by ~20 percentage points over a baseline and reaching ~94% with minimal manual verification ⁵¹ ⁵².", IEEE Transactions on Software Engineering ⁴⁹

Semantic web-based automated compliance checking with integration of Finite Element analysis ⁵³, Panagiotis Patlakas; Ioannis P. Christovasilis; Lorenzo Riparbelli; Franco K. T. Cheung; Edlira Vakaj, 2024 ⁵⁴, 10.1016/j.aei.2024.102448 ⁵⁵,"Automatic Compliance Checking (ACC) is a promising approach for ensuring building and planning regulations compliance, especially with BIM/IFC. However, computational analysis methods like Finite Element Analysis (FEA) have seen little integration in ACC. This work presents a holistic approach that embeds FEA results into IFC and uses a Semantic Web-based system for ACC. A top-level Building Compliance Ontology (BCO) is developed to represent regulatory statements (not just building components). The approach is evaluated on a modular steel structure's code compliance: an open-source FEA solver provides results which are integrated via middleware, BCO is extended for structural engineering, and compliance is automatically checked with SHACL scripts in a Semantic Web environment ⁵⁶ ⁵⁷.", Advanced Engineering Informatics ⁵⁴

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